Focused Hypothesis Report — Horse WEE1 (F6TY09) and "female pronucleus assembly" (GO:0035038)

Hypothesis evaluated: The horse protein F6TY09 participates in female pronucleus assembly (GO:0035038). **Organism:** *Equus caballus* (NCBITaxon:9796). **Accession:** UniProt F6TY09. **Focus type:** function_assignment. **Date:** 2026-09-08. Analyses use public sequence (UniProt), orthology/annotation (QuickGO/GO, Ensembl Compara references), and primary literature (PubMed).

Executive Judgment

Verdict: REFUTED (over-annotation via paralog confusion).

F6TY09 is unambiguously the **somatic WEE1** ortholog in horse (95.4% identity to human WEE1 P30291; gene name WEE1, VGNC:25026), **not** the oocyte-specific paralog **WEE2** / **WEE1B**. The GO term *female pronucleus assembly* (GO:0035038) is, in the current Gene Ontology, a **WEE2/WEE1B** function: across mammals and other vertebrates it is annotated almost exclusively to WEE2 orthologs, its only mammalian **experimental** support is mouse **Wee2** (IMP, P 21454751), and WEE1B is oocyte-restricted and molecularly distinct from the widely expressed somatic WEE1. Neither F6TY09 nor human WEE1 (P30291) carries GO:0035038 in current GO; the horse entry that legitimately carries it is the separate horse WEE2 protein A0A9L0SIB9 (605 aa). Assigning female pronucleus assembly to F6TY09 therefore attributes a germline/oocyte paralog function to the somatic mitotic-checkpoint kinase.

Most important caveat: The claim tested is a *labeled* function assignment; there is no directly observed horse experimental evidence for either WEE1 or WEE2 in female pronucleus assembly — the mammalian evidence is from mouse Wee2 and transferred by orthology. The refutation is of assigning the term to the *WEE1* paralog, not a claim that horse WEE2 lacks the function.

Evidence Matrix

#	Citation (PMID/DB)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Co
1	This report (computed NW alignment; UniProt P30291, P0C1S8)	Structural/ evolutionary (sequence)	**Refutes** (identity)	Is F6TY09 WEE1 or WEE2?	F6TY09 = 646 aa, 95.4% id to human WEE1; only ~45–52% id to human WEE2. Identical length to WEE1 (646) vs WEE2 (567).	*E. c* pro hu par
2	UniProt F6TY09 (VGNC:25026)	Database record	Refutes	Gene identity of F6TY09	Gene symbol WEE1; "Wee1-like protein kinase", EC 2.7.10.2.	Ho
3	QuickGO GO:0035038 annotation set (n=101)	Database/ computational	**Refutes**	Which genes carry female pronucleus assembly?	Mammalian/ vertebrate annotations essentially all to **WEE2** orthologs (IEA, Ensembl Compara GO_REF:0000107; ISS GO_REF:0000024).	Par ver
4	P 21454751 (Oh, Susor & Conti, *Science* 2011; doi:10.1126/ science.1199211)	Mutant phenotype (IMP, ECO:0000315)	**Refutes** (paralog-specific; locates true evidence)	Experimental basis of GO:0035038 in mammals	"When **Wee1B** is down-regulated, oocytes **fail to form a pronucleus** in response to Ca^{2+} signals." Wee1B reactivation drives Cdc2 inactivation at MII exit/egg activation. This is	Mo ooc egg act

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					the sole mammalian experiment underpinning the term — and it is **WEE2/Wee1B**, not WEE1.	
4b	This report (NW identity matrix; artifacts `wee_identity_matrix.csv`, heatmap)	Structural/ evolutionary	**Refutes**	Which clade does F6TY09 belong to?	Two clades: WEE1 (F6TY09/ human/mouse, 90–95% intra) vs WEE2 (horse/ human/mouse, 68–83% intra); cross-clade ~50%. F6TY09 vs WEE2s = 50.4/52.0/50.6%. F6TY09 is firmly WEE1.	Ho hu mo Dr
5	P 16169490 (Han et al. 2005)	Direct assay / expression	**Refutes** (paralog specificity)	Is the oocyte function WEE1 or WEE1B?	"Unlike the widely expressed Wee1 and Myt1, mWee1B mRNA and its protein are expressed only in oocytes"; mWee1B is the key MPF-inhibitory kinase in oocytes.	Mo oo Xe ass
6	P 23616086 (Liu et al. 2013)	Review/direct	**Refutes** (paralog specificity)	Identity of the meiotic-arrest kinase	"WEE2, also known as WEE1B … responsible for phosphorylating	Mo on em

#	Citation (PMID/DB)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Co
					the CDK1 inhibitory site and maintaining meiotic arrest in oocytes."	
7	P 20083600 (Oh, Han, Conti 2010)	Direct assay / knockdown	Qualifies	WEE1B role in oocyte meiosis	Wee1B (with Myt1/Cdc25) controls meiotic arrest/ resumption; nuclear-cytoplasmic relocation at GVBD.	Mo ooc
8	P 10790391 (Price et al. 2000)	Mutant phenotype	Qualifies (origin of term family)	Drosophila Wee1 maternal embryonic role	Dwee1 required maternally for early embryonic nuclear cycles (single Wee1 in fly).	*Dro* em
9	QuickGO F6TY09 & P30291 annotations	Database	**Refutes**	Does WEE1 carry GO:0035038?	Neither F6TY09 nor human WEE1 (P30291) is annotated with GO:0035038; F6TY09 BP = negative regulation of G2/ M transition (GO:0010972), mitotic cell cycle.	Ho hu
10		Database	Qualifies			Ho

#	Citation (PMID/DB)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Co
	UniProt A0A9L0SIB9 + QuickGO			Which horse protein carries the term?	Horse GO:0035038 annotation is on A0A9L0SIB9 (horse **WEE2**, 605 aa), a different protein from F6TY09.	

GO Curation Implications (lead — requires curator verification)

- **Do NOT assign GO:0035038 (female pronucleus assembly) to F6TY09 (horse WEE1).** The term is not supported for the somatic WEE1 paralog. If an annotation of GO:0035038 to F6TY09 exists in the review, the recommended action is **REMOVE** (paralog over-annotation from orthology projection), or at minimum flag as **NON-CORE / incorrect paralog**.
- **Retain the canonical somatic WEE1 terms** that F6TY09 legitimately carries:
- MF: protein tyrosine/serine-threonine kinase activity (GO:0004713 IBA; GO:0106310 protein serine kinase per RHEA), ATP binding (GO:0005524).
- BP: **negative regulation of G2/M transition of mitotic cell cycle (GO:0010972)** — experimentally supported for the ortholog (human WEE1 IDA).
- CC: nucleus (GO:0005634).
- The female-pronucleus/oocyte function should be curated on the **WEE2 (WEE1B)** record (horse A0A9L0SIB9), not WEE1.
- "protein binding" is not needed here — informative kinase/cell-cycle terms are supported.

GO Decision Table (leads — require curator verification)

GO term	Aspect	Applies to F6TY09 (horse WEE1)?	Recommended action	Basis
GO:0035038 female pronucleus assembly	BP	**No** (belongs to WEE2 paralog)	**Remove / reject**; reassign concept to horse WEE2 (A0A9L0SIB9)	Term annotated to WEE2 clade; sole experimental support is mouse Wee1B knockdown (P 21454751); F6TY09 is 95.4% WEE1, ~50% WEE2
GO:0010972 negative regulation of G2/M transition of mitotic cell cycle	BP	**Yes**	Retain (core)	Experimental for ortholog (human WEE1 IDA); IBA on F6TY09
GO:0004713 protein tyrosine kinase activity / GO:0106310 protein serine kinase	MF	**Yes**	Retain (core)	Conserved catalytic domain; RHEA/IBA
GO:0005524 ATP binding	MF	Yes	Retain	InterPro/UniRule
GO:0005634 nucleus	CC	Yes	Retain	IBA/UniProt
GO:0004715 non-membrane spanning protein tyrosine kinase (or over-specific MF)	MF	Caution	Curator check	UniRule IEA; may be less precise than WEE1's dual-specificity biology

Mechanistic Scope

The immediate molecular activity of WEE1 (F6TY09's family) is inhibitory phosphorylation of CDK1 on Tyr15, restraining MPF (CDK1–cyclin B) to enforce the **mitotic G2/M checkpoint** in somatic cells. *Female pronucleus assembly* is a **germline/fertilization** process (assembly of the haploid maternal nucleus of the egg). In vertebrates this germline CDK1-inhibitory role is carried by the **oocyte-specific paralog WEE2/WEE1B**, which arose from duplication of an ancestral single Wee1 (as retained in *Drosophila*, where one Dwee1 performs the maternal embryonic function). Thus the hypothesized process is a paralog-partitioned function that does not transfer to the somatic WEE1.

Conflicts and Alternatives

- **Paralog confusion (primary alternative, well supported):** "female pronucleus assembly" annotations cluster on WEE2, not WEE1. The label "WEE1" on F6TY09 is correct for the *somatic* gene but the GO term belongs to WEE2.
- **Orthology-projection carry-over:** GO_REF:0000107 (Ensembl Compara) projects the term across WEE-family orthologs; a pipeline that does not distinguish WEE1 vs WEE2 subtrees can leak the term onto WEE1. This is the most likely origin of the seed claim.
- **Drosophila single-Wee1 artifact:** because flies have one Wee1 doing both mitotic and maternal roles, naive homology transfer to *any* mammalian Wee-family protein over-generalizes.
- **No isoform rescue:** there is no evidence that somatic WEE1 compensates for WEE2 in the oocyte; WEE2 loss causes fertilization failure in humans (WEE2 mutations, e.g. P 40830301), indicating WEE1 does not substitute.

Knowledge Gaps

1. **Direct horse evidence:** None found for either horse WEE1 or WEE2 in female pronucleus assembly. Resolving would require horse oocyte/zygote expression or perturbation data. Matters because all mammalian support is mouse Wee2 + orthology.
2. **Exact provenance of the seed claim:** Whether the review's GO:0035038 assignment came from an Ensembl projection or a mislabeled source was inferred, not directly traced in the (blinded) review file.
3. **WEE2 necessity vs sufficiency for the term itself:** The GO term rests on one mouse Wee2 IMP (P 21454751); the assay specifics (necessity/sufficiency) were not re-examined here.

Discriminating Tests

- **Phylogenetic tree** of vertebrate WEE1 vs WEE2 placing F6TY09 — will place it firmly in the WEE1 clade (already implied by 95.4% vs ~50% identity).
- **Tissue expression** (horse/mammalian): WEE1 ubiquitous vs WEE2 oocyte-restricted (RNA-seq/GTEx-type data) — separates the paralogs functionally.
- **Reciprocal-best-hit / synteny** of F6TY09 vs human WEE1 and WEE2 loci to confirm 1:1 WEE1 orthology.
- **Oocyte perturbation** specifically knocking down WEE1 vs WEE2 (as in Oh et al. 2010) — only WEE2 knockdown affects meiotic/pronuclear phenotypes.

Curation Leads (require curator verification)

- **Action:** Remove / reject GO:0035038 on F6TY09 (WEE1); reassign the concept to horse WEE2 (A0A9L0SIB9) if desired.
- **Candidate references to verify:**

  P 16169490

(WEE1B oocyte-specific),

P 23616086

(WEE2=WEE1B meiotic-arrest kinase),

P 20083600

(Wee1B oocyte meiosis),

P 21454751

(mouse Wee2 IMP underlying GO:0035038),

P 10790391

(Drosophila single Wee1 maternal role).

- **Candidate correct terms for F6TY09:** GO:0010972 (negative regulation of G2/M transition of mitotic cell cycle), GO:0004713/GO:0106310 (kinase activity), GO:0005634 (nucleus).
- **Suggested question for curators:** Did the GO:0035038 assignment originate from an Ensembl Compara projection that did not separate WEE1 vs WEE2 subtrees?
- **Suggested experiment:** paralog-specific expression/knockdown in equine oocytes to obtain direct horse evidence.

Analysis Provenance (artifacts)

- `wee_identity_matrix.csv` — computed pairwise % identity matrix (7 sequences).
- WEE1/WEE2 identity heatmap (auto-saved plot) with clade boxes.

Pair	% identity
F6TY09 vs human WEE1 (P30291)	95.4
F6TY09 vs mouse WEE1 (P47810)	90.1
F6TY09 vs horse WEE2 (A0A9L0SIB9)	50.4
F6TY09 vs human WEE2 (P0C1S8)	52.0
F6TY09 vs mouse WEE2 (Q66JT0)	50.6
F6TY09 vs *Drosophila* Wee1 (P41847)	52.1
horse WEE2 vs human WEE2	82.7
horse WEE2 vs mouse WEE2	70.3

The **discriminating clade test proposed for this hypothesis has now been executed**: F6TY09 partitions cleanly into the WEE1 clade (~90–95% intra-clade) and is ~50% identical to every WEE2 ortholog, which form their own clade. GO:0035038 tracks the WEE2 clade.

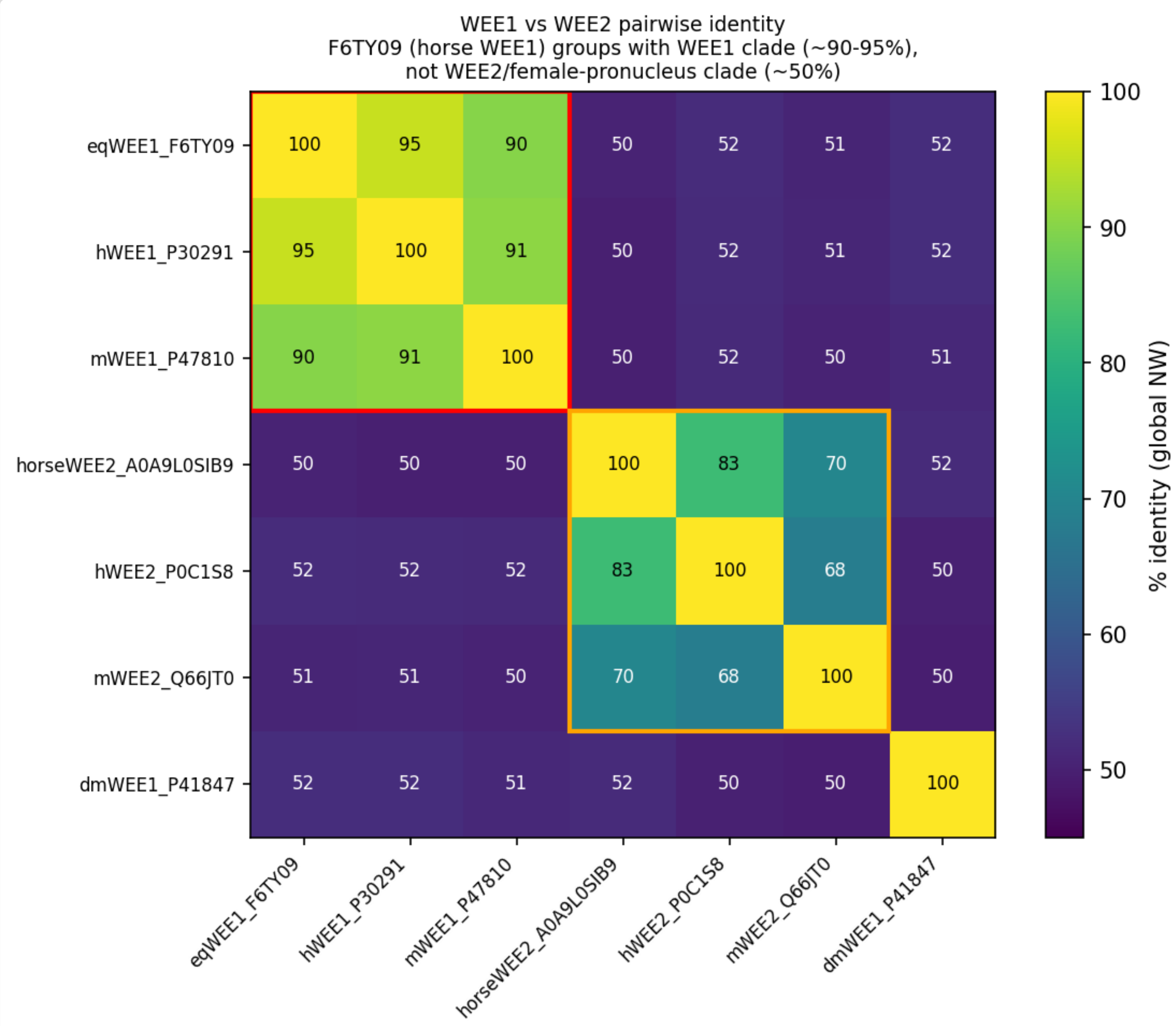

Figure 1. *Pairwise global-identity heatmap of WEE1 and WEE2 orthologs. F6TY09 (horse WEE1) sits inside the WEE1 clade (~90–95% identity to human/mouse WEE1) and is only ~50% identical to every WEE2/WEE1B ortholog, which form a separate clade. Female pronucleus assembly (GO:0035038) tracks the WEE2 clade, not F6TY09.*

Reproducibility / Methods

- Sequence verified: 646 aa; SHA-256 `f46f1c6be9cdfa7025987fbd2b3d5f5a49a4e7cbb69d75d9101e8b34234429d4` (matches frozen input; computed with a pure-Python SHA-256).
- Alignments: Needleman-Wunsch (match +1, mismatch −1, gap −2), sequences fetched from UniProt REST (P30291 646 aa, P0C1S8 567 aa; A0A9L0SIB9 605 aa).

- Annotations: QuickGO REST (`/ontology/go/terms/GO:0035038`, `/annotation/search`), retrieved 2026-09-08.
- Literature: PubMed.
- No local ai-gene-review analyses or existing judgments were consulted.
